

# Quadratic functions - Set 1 - Solutions

Specification MM1.3, M4.5, M4.11-M4.14.

## Answer key

| Question | Answer | Topic               |
|----------|--------|---------------------|
| 1        | E      | Quadratic functions |
| 2        | B      | Quadratic functions |
| 3        | A      | Quadratic functions |
| 4        | C      | Quadratic functions |
| 5        | B      | Quadratic functions |
| 6        | C      | Quadratic functions |
| 7        | D      | Quadratic functions |
| 8        | B      | Quadratic functions |
| 9        | C      | Quadratic functions |
| 10       | D      | Quadratic functions |
| 11       | C      | Quadratic functions |
| 12       | C      | Quadratic functions |

## Worked solutions

### Question 1 QUA-01 · Quadratic functions · spec MM1.3

The equation  $x^2 + (k - 2)x + (k + 1) = 0$  has two distinct real roots. Find the complete set of possible values of  $k$ .

**Answer: E**

Discriminant  $(k - 2)^2 - 4(k + 1) = k^2 - 8k = k(k - 8) > 0$ , so  $k < 0$  or  $k > 8$ .

#### Full method - how and why

'Two distinct real roots' is always a discriminant condition. The unknown is in the coefficients, so the discriminant becomes a quadratic **in  $k$**  and you finish with an inequality.

- 1 Identify  $a = 1$ ,  $b = k - 2$ ,  $c = k + 1$  and write the discriminant:  $(k - 2)^2 - 4(k + 1)$ .  
why: 'Two distinct real roots'  $\Leftrightarrow b^2 - 4ac > 0$ . Everything else in the question is decoration.
- 2 Expand and simplify:  $k^2 - 4k + 4 - 4k - 4 = k^2 - 8k$ .  
why: The constants cancel, which is the sign you have expanded correctly.
- 3 Factorise and solve  $k(k - 8) > 0$ .  
why: A quadratic inequality needs its roots and its shape: roots 0 and 8, opening upwards.
- 4 The upward parabola is positive outside its roots:  $k < 0$  or  $k > 8$ .  
why: Sketch it if in doubt - 'outside the roots' for  $> 0$ , 'between the roots' for  $< 0$ .

#### Where it goes wrong

- Expanding  $(k - 2)^2$  as  $k^2 - 4$ , which gives  $k^2 - 4k - 8$  and a different answer set.
- Reading the sign chart backwards and choosing  $0 < k < 8$  - that is where the discriminant is **negative** and there are no real roots.

**Takeaway** Conditions on the number of roots are conditions on  $b^2 - 4ac$ ; if the coefficients contain a parameter, you end up solving a quadratic inequality in that parameter.

### Question 2 QUA-02 · Quadratic functions · spec MM1.3

By completing the square, find the minimum value of  $2x^2 - 12x + 23$ .

**Answer: B**

$2x^2 - 12x + 23 = 2(x - 3)^2 - 18 + 23 = 2(x - 3)^2 + 5$ , so the minimum is 5 (at  $x = 3$ ).

#### Full method - how and why

Minimum value of a quadratic: complete the square. The leading coefficient 2 must be factored out **before** completing the square.

- 1 Factor the 2 out of the  $x$  terms:  $2(x^2 - 6x) + 23$ .  
why: Completing the square only works on  $x^2 + (\text{something})x$ ; the coefficient of  $x^2$  must be 1 inside the bracket.
- 2 Complete the square inside:  $x^2 - 6x = (x - 3)^2 - 9$ , so the expression is  $2[(x - 3)^2 - 9] + 23$ .  
why: Halve the coefficient of  $x$  ( $-6 \rightarrow -3$ ) and subtract the square of that half.
- 3 Multiply out the 2:  $2(x - 3)^2 - 18 + 23 = 2(x - 3)^2 + 5$ .  
why: The  $-9$  is inside the bracket, so it is multiplied by 2 as well - this is where marks are lost.
- 4  $2(x - 3)^2 \geq 0$ , so the least value is 5, at  $x = 3$ .  
why: A square is never negative, and it is zero exactly at the vertex.

#### Where it goes wrong

- Forgetting to multiply the  $-9$  by 2, giving  $2(x - 3)^2 + 14$  and the answer 14.
- Quoting the  $x$ -coordinate 3 instead of the minimum value 5 - both are offered.

**Takeaway** Take the leading coefficient outside first, complete the square, then multiply back in; the constant left outside is the minimum.

### Question 3 QUA-03 · Quadratic functions · spec MM1.3

The quadratic  $y = ax^2 + bx + c$  has its vertex at  $(2, -3)$  and passes through  $(0, 5)$ . Find  $a + b + c$ .

**Answer: A**

$y = a(x - 2)^2 - 3$ ; at  $x = 0$ :  $4a - 3 = 5$ , so  $a = 2$ . Then  $y = 2(x - 2)^2 - 3 = 2x^2 - 8x + 5$ , and  $a + b + c = 2 - 8 + 5 = -1$  (which is just  $y(1)$ ).

#### Full method - how and why

Given a vertex, the completed-square form is the natural one. The clever final step is noticing that  $a + b + c$  is just  $y(1)$ .

- 1 Write the completed-square form using the vertex:  $y = a(x - 2)^2 - 3$ .  
*why:* The vertex  $(h, k)$  appears directly in  $y = a(x - h)^2 + k$ , so two of the three unknowns are already known.
- 2 Use the point  $(0, 5)$ :  $a(0 - 2)^2 - 3 = 5 \Rightarrow 4a = 8 \Rightarrow a = 2$ .  
*why:* One extra point is exactly what is needed to pin down the remaining constant.
- 3 Note that  $a + b + c$  is the value of  $ax^2 + bx + c$  at  $x = 1$ .  
*why:* Substituting  $x = 1$  makes every power of  $x$  equal to 1 - a standard shortcut worth remembering.
- 4 Evaluate:  $y(1) = 2(1 - 2)^2 - 3 = 2 - 3 = -1$ .  
*why:* No need to expand into  $2x^2 - 8x + 5$  at all.

#### Where it goes wrong

- Expanding and then mis-collecting:  $2(x - 2)^2 - 3 = 2x^2 - 8x + 5$ , and slips in the  $-8$  give 3 or 7.
- Using the vertex as  $(-2, -3)$  - in  $a(x - h)^2 + k$  the sign of  $h$  is already flipped.

**Takeaway**  $f(1)$  is the sum of the coefficients; and a given vertex means you should start from  $a(x - h)^2 + k$ .

### Question 4 QUA-05 · Quadratic functions · spec MM1.3

For what value of  $c$  does the line  $y = 2x + c$  touch the parabola  $y = x^2 - 4x + 7$ ?

**Answer: C**

Equate:  $x^2 - 6x + (7 - c) = 0$ . Tangency needs discriminant zero:  $36 - 4(7 - c) = 0$ , so  $8 + 4c = 0$  and  $c = -2$ .

#### Full method - how and why

'Touches' means exactly one intersection, which after substitution means a repeated root: discriminant zero.

- 1 Set the two expressions equal:  $2x + c = x^2 - 4x + 7$ .  
*why:* Intersections of  $y = f(x)$  and  $y = g(x)$  are the solutions of  $f(x) = g(x)$ .
- 2 Collect into a standard quadratic:  $x^2 - 6x + (7 - c) = 0$ .  
*why:* Everything to one side, so the discriminant test can be applied.
- 3 Set the discriminant to zero:  $(-6)^2 - 4(1)(7 - c) = 0$ .  
*why:* Touching means a repeated root, i.e. the line meets the curve once - exactly the  $b^2 = 4ac$  case.
- 4 Solve:  $36 - 28 + 4c = 0 \Rightarrow 4c = -8 \Rightarrow c = -2$ .  
*why:* Watch the double negative:  $-4(7 - c) = -28 + 4c$ .

#### Where it goes wrong

- Sign slip in  $-4(7 - c)$  giving  $c = 2$ , which is offered.
- Using  $b^2 - 4ac > 0$  (crossing) rather than  $= 0$  (touching).

**Takeaway** Tangency between a line and a curve: substitute, then set the discriminant of the resulting quadratic to zero.

### Question 5 QUA-07 · Quadratic functions · spec MM1.3

How many real solutions does the equation  $x^4 - 5x^2 + 4 = 0$  have?

**Answer: B**

Let  $u = x^2$ :  $u^2 - 5u + 4 = (u - 1)(u - 4) = 0$ , so  $x^2 = 1$  or  $x^2 = 4$ , giving  $x = \pm 1, \pm 2$ : four solutions.

#### Full method - how and why

A quartic that is quadratic in  $x^2$ . The trap is counting the solutions of the substituted equation rather than of the original.

1 Substitute  $u = x^2$ :  $u^2 - 5u + 4 = 0$ .

why: Only even powers of  $x$  appear, which is what makes the substitution work.

2 Factorise:  $(u - 1)(u - 4) = 0$ , so  $u = 1$  or  $u = 4$ .

why: Two values of  $u$  - but  $u$  is not the answer.

3 Undo the substitution:  $x^2 = 1$  gives  $x = \pm 1$ ;  $x^2 = 4$  gives  $x = \pm 2$ .

why: Each **positive** value of  $u$  produces **two** values of  $x$ ; a negative value would produce none.

4 Count: four real solutions.

why: The question asks how many, not what they are.

#### Where it goes wrong

- Answering 2 because the quadratic in  $u$  has two roots.
- Forgetting the negative square roots and answering 2 again.

**Takeaway** After a substitution, always translate back: each positive value of  $x^2$  gives two values of  $x$ , zero gives one, negative gives none.

### Question 6 QUA-08 · Quadratic functions · spec MM1.3

The equation  $kx^2 + 4x + k = 0$  has no real roots. Find the complete set of values of  $k$ .

**Answer: C**

Need  $16 - 4k^2 < 0$ , i.e.  $k^2 > 4$ , so  $k < -2$  or  $k > 2$ . (If  $k = 0$  the equation is linear with one root, so  $k = 0$  is excluded anyway.)

#### Full method - how and why

'No real roots' is the discriminant  $< 0$ . The extra subtlety is the coefficient  $k$  appearing in the  $x^2$  term, so  $k = 0$  needs a thought.

1 Write the discriminant:  $4^2 - 4(k)(k) = 16 - 4k^2$ .

why: Here  $a = k$  and  $c = k$  - both the first and last coefficients are the parameter.

2 Set it negative:  $16 - 4k^2 < 0 \Rightarrow k^2 > 4$ .

why: No real roots  $\Leftrightarrow b^2 - 4ac < 0$ .

3 Solve  $k^2 > 4$ :  $k < -2$  or  $k > 2$ .

why:  $k^2 > 4$  means  $|k| > 2$ , which is two separate intervals, not  $-2 < k < 2$ .

4 Check  $k = 0$  separately: the equation becomes  $4x = 0$ , which has the root  $x = 0$ .

why: If  $k = 0$  it is not a quadratic at all, so the discriminant test does not apply - but  $k = 0$  is excluded by the answer anyway.

#### Where it goes wrong

- Solving  $k^2 > 4$  as  $-2 < k < 2$  - that is the solution of  $k^2 < 4$ .
- Ignoring the possibility  $k = 0$  entirely (harmless here, fatal in questions where 0 lies inside the answer set).

**Takeaway**  $k^2 > c$  gives two outer intervals; and whenever the leading coefficient contains the parameter, check the value that makes it zero.

### Question 7 QUA-10 · Quadratic functions · spec MM1.3

The quadratic  $x^2 + px + q$  has minimum value  $-4$  and one of its roots is 0. Which of the following could be the quadratic?

**Answer: D**

Minimum  $-4$  means  $x^2 + px + q = (x - h)^2 - 4$ , with roots  $h \pm 2$ . One root is 0, so  $h = \pm 2$ : the quadratic is  $x^2 - 4x$  or  $x^2 + 4x$ . Only  $x^2 - 4x$  is offered.

#### Full method - how and why

Two pieces of information about a quadratic - its minimum and one root - pin the whole thing down. Working in completed-square form makes the roots visible.

- 1 Minimum  $-4$  and leading coefficient 1 means  $x^2 + px + q = (x - h)^2 - 4$  for some  $h$ .  
why: A monic quadratic with minimum  $m$  must be  $(x - h)^2 + m$ ; only the position  $h$  is unknown.
- 2 Its roots satisfy  $(x - h)^2 = 4$ , i.e.  $x = h \pm 2$ .  
why: Solve the completed-square form directly rather than expanding.
- 3 One root is 0, so  $h + 2 = 0$  or  $h - 2 = 0$ :  $h = -2$  or  $h = 2$ .  
why: Both cases are legitimate; the options decide which one is wanted.
- 4 That gives  $x^2 + 4x$  or  $x^2 - 4x$ ; the list offers  $x^2 - 4x$ .  
why: Check it:  $x^2 - 4x = (x - 2)^2 - 4$ , minimum  $-4$ , roots 0 and 4.

#### Where it goes wrong

- Picking  $x^2 - 2x - 3$ : its minimum is  $-4$  but its roots are  $-1$  and  $3$ , so 0 is not a root.
- Checking only the minimum (several options have minimum  $-4$ ) or only the roots.

**Takeaway** Test every stated condition - an option that satisfies one of them is designed to catch you.

### Question 8 QUA-11 · Quadratic functions · spec MM1.3

The curve  $y = x^2 - 2ax + a^2 + a - 6$  has its vertex on the  $x$ -axis. Find the value of  $a$ .

**Answer: B**

Complete the square:  $y = (x - a)^2 + a - 6$ . The vertex is  $(a, a - 6)$ , which lies on the  $x$ -axis when  $a = 6$ .

#### Full method - how and why

'Vertex on the  $x$ -axis' means the minimum value is zero, i.e. the quadratic has a repeated root. Completing the square is quicker than the discriminant here.

- 1 Complete the square in  $x$ , treating  $a$  as a constant:  $x^2 - 2ax + a^2 = (x - a)^2$ , so  $y = (x - a)^2 + (a - 6)$ .  
why: The first three terms are already a perfect square - that is why the question is built with  $a^2$  in it.
- 2 Read off the vertex:  $(a, a - 6)$ .  
why: In  $(x - h)^2 + k$  the vertex is  $(h, k)$ .
- 3 On the  $x$ -axis means the  $y$ -coordinate is zero:  $a - 6 = 0$ , so  $a = 6$ .  
why: 'On the  $x$ -axis' is a statement about  $y$ , not  $x$  - the commonest misread in this question.

#### Where it goes wrong

- Setting the  $x$ -coordinate to zero and answering  $a = 0$ .
- Using the discriminant but mis-expanding  $4a^2 - 4(a^2 + a - 6)$ , which should collapse to  $-4(a - 6)$ .

**Takeaway** Vertex on the  $x$ -axis  $\Leftrightarrow$  repeated root  $\Leftrightarrow$  discriminant zero - three descriptions of the same situation.

### Question 9 QUA-12 · Quadratic functions · spec MM1.3

The roots of  $2x^2 + bx + 18 = 0$  are equal. Find the sum of the possible values of  $b$ .

**Answer: C**

$$b^2 - 144 = 0, \text{ so } b = \pm 12; \text{ their sum is } 0.$$

#### Full method - how and why

'Equal roots' is the discriminant-zero condition; the twist is that it gives two values of  $b$  and the question wants their sum.

- 1 Write the discriminant with  $a = 2$ ,  $c = 18$ :  $b^2 - 4(2)(18) = b^2 - 144$ .

why: Substitute carefully -  $4ac$  is 144, not 36.

- 2 Set it to zero:  $b^2 = 144$ , so  $b = 12$  or  $b = -12$ .

why: Equal (repeated) roots means the discriminant vanishes; a square equal to a positive number always has two solutions.

- 3 Add them:  $12 + (-12) = 0$ .

why: The two values are symmetric about zero, so the sum is 0 without any further work.

#### Where it goes wrong

- Giving only the positive root and answering 12.
- Reading 'sum of the possible values of  $b$ ' as 'the value of  $b$ '.

**Takeaway**  $b^2 = k$  always has two solutions  $\pm\sqrt{k}$ ; read whether the question wants one of them, both, or a combination.

### Question 10 QUA-13 · Quadratic functions · spec MM1.3

Which of the following quadratics has its vertex in the second quadrant (where  $x < 0$  and  $y > 0$ )?

**Answer: D**

Complete the square: A vertex  $(-2, -3)$ ; B  $(2, 1)$ ; C  $(-1, -2)$ ; D  $(-3, 1)$ , which has  $x < 0$  and  $y > 0$ ; E  $(1, 2)$ . So D.

#### Full method - how and why

A 'which one' question about vertices. Complete the square on each option - but use the sign of the  $x$ -coordinate to eliminate most of them first.

- 1 Recall that for  $y = x^2 + bx + c$  the vertex is at  $x = -\frac{b}{2}$ .

why: You only need the vertex position, not the whole completed square, to test the first condition  $x < 0$ .

- 2 Second quadrant needs  $x < 0$ , so  $-\frac{b}{2} < 0$ , i.e.  $b > 0$ : only  $y = x^2 + 4x + 1$  and  $y = x^2 + 6x + 10$  survive among the upward parabolas.

why: Eliminating on one condition first halves the work.

- 3 Compute the  $y$ -coordinates:  $x^2 + 4x + 1 = (x + 2)^2 - 3$ , vertex  $(-2, -3)$ : fails  $y > 0$ .  $x^2 + 6x + 10 = (x + 3)^2 + 1$ , vertex  $(-3, 1)$ : passes.

why: Now apply the second condition to the short list.

- 4 Check the downward ones quickly:  $-x^2 - 2x - 3$  has vertex  $(-1, -2)$  and  $-x^2 + 2x + 1$  has vertex  $(1, 2)$  - neither is in the second quadrant.

why: Worth 10 seconds of confirmation on a 'which one' question.

#### Where it goes wrong

- Reading the vertex of  $(x + 3)^2 + 1$  as  $(3, 1)$  - the sign flips.
- Mixing up the quadrants: the second quadrant is up and to the left.

**Takeaway** On 'which of these' questions, test the cheapest condition first and eliminate before doing any heavy algebra.

**Question 11** QUA-15 · Quadratic functions · spec MM1.3

The graph of  $y = x^2 + bx + c$  crosses the  $x$ -axis at  $(1, 0)$  and  $(5, 0)$ . What is the  $y$ -coordinate of the vertex?

**Answer: C**

$$y = (x - 1)(x - 5), \text{ vertex at } x = 3: y = (2)(-2) = -4.$$

**Full method - how and why**

Given the two roots, use the factorised form and the symmetry of the parabola - no expansion needed.

- 1 Write the factorised form:  $y = (x - 1)(x - 5)$ .

why: Roots at 1 and 5 with leading coefficient 1 (given, since the equation is  $x^2 + bx + c$ ).

- 2 The vertex lies midway between the roots:  $x = \frac{1+5}{2} = 3$ .

why: A parabola is symmetric, so its turning point sits on the axis of symmetry, exactly halfway between the roots.

- 3 Evaluate there:  $y = (3 - 1)(3 - 5) = 2 \times (-2) = -4$ .

why: Substituting into the factorised form is far quicker than expanding to  $x^2 - 6x + 5$ .

**Where it goes wrong**

- Averaging the roots but then substituting into the wrong expression, e.g. getting  $-9$  from  $(3)^2 - 6(3) + 5$  miscomputed.
- Giving the  $x$ -coordinate 3 instead of the  $y$ -coordinate.

**Takeaway** Midpoint of the roots gives the axis of symmetry; substitute into the factorised form for the vertex height.

**Question 12** QUA-16 · Quadratic functions · spec MM1.3

Given that  $x^2 + y^2 = 10$  and  $x + y = 4$ , find the value of  $xy$ .

**Answer: C**

$$(x + y)^2 = x^2 + y^2 + 2xy, \text{ so } 16 = 10 + 2xy \text{ and } xy = 3.$$

**Full method - how and why**

A symmetric pair of equations. Expanding  $(x + y)^2$  links the two given quantities to  $xy$  directly.

- 1 Write the identity  $(x + y)^2 = x^2 + y^2 + 2xy$ .

why: This is the only identity that connects a sum, a sum of squares and a product.

- 2 Substitute the given values:  $4^2 = 10 + 2xy$ .

why:  $x + y = 4$  and  $x^2 + y^2 = 10$  are both given, so only  $xy$  is unknown.

- 3 Solve:  $16 - 10 = 2xy \Rightarrow xy = 3$ .

why: Halve at the end - forgetting to divide by 2 gives 6, which is offered.

**Where it goes wrong**

- Answering 6 by omitting the factor 2 in  $2xy$ .
- Trying to solve for  $x$  and  $y$  individually (they are 1 and 3) - slower and unnecessary.

**Takeaway**  $(x + y)^2 = x^2 + y^2 + 2xy$  converts between the three symmetric quantities; recognise which two you are given.